

Javier Romero

Curriculum Vitae

Contact

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Research Interests

My research interest lies in the intersection between graphics and vision. I am interested in image generation for the purpose of both understanding the world through vision and presenting new visual experiences to customers.

Professional Experience

I have worked on human estimation, modeling and generation for the last fifteen years. During my academic years, I have performed basic research on graphics and vision with the goal of teaching computers how to perceive people's shape and motion. As a result, I published highly influential articles on body models like [SMPL](#), [FLAME](#), [MANO](#) or [MHR](#), among others. In Amazon, I worked as an individual contributor and science lead in projects that use body estimation to create customized garments like [Made for You](#). I have also contributed to fitness projects like [Halo](#) or new shopping experiences like Amazon Go, among others. In my last position at Meta, I work on human data collection and modelling with the goal of achieving VR communication indistinguishable from reality.

Education

2007–2011 *PhD in Computer Science,*
Kungliga Tekniska Högskolan(KTH)
Advisor: Prof. Danica Kragic

2001-2006 *M.S. Telecommunication Engineering,*
School of Telecommunication Engineering (ETSIT)
in Technical University of Madrid (UPM)
GPA: 8.23/10.00
Thesis: *Binocular Hand Tracking and Pose Estimation*
Kungliga Tekniska Högskolan (KTH)
Stockholm, Sweden
Adviser: Prof. Jan-Olof Eklundh, Docent Danica Kragic

Employment

7/2021–present Research Scientist Lead
Meta
Madrid, Spain

10/2020–7/2021 Principal Applied Scientist
Amazon
Barcelona, Spain

10/2017–9/2020	Senior Applied Scientist Amazon Barcelona, Spain
2/2017–10/2017	Senior CV/ML Research Scientist Body Labs Barcelona, Spain
6/2014–1/2017	Research Scientist MPI for Intelligent Systems, Perceiving Systems Department Tübingen, Germany
1/2012–6/2014	Postdoctoral Researcher MPI for Intelligent Systems, Perceiving Systems Department Tübingen, Germany

Selected list of publications

Yuxuan Xue, Ruofan Liang, Egor Zakharov, Timur M. Bagautdinov, Chen Cao, Giljoo Nam, Shunsuke Saito, Gerard Pons-Moll, and **Javier Romero**. Georelight: Learning joint geometrical relighting and reconstruction with flexible multi-modal diffusion transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2026, Denver, CO, USA, June 3-7, 2026*, pages 29771–29780. Computer Vision Foundation, 2026.

Jinhyung Park, **Javier Romero**, Shunsuke Saito, Fabian Prada, Takaaki Shiratori, Yichen Xu, Federica Bogo, Shouo-I Yu, Kris Kitani, and Rawal Khrodkar. ATLAS: decoupling skeletal and shape parameters for expressive parametric human modeling. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 6508–6518. IEEE, 2025.

Tobias Kirschstein, **Javier Romero**, Artem Sevastopolsky, Matthias Nießner, and Shunsuke Saito. Avat3r: Large animatable gaussian reconstruction model for high-fidelity 3d head avatars. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 12089–12100. IEEE, 2025.

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Sergey Prokudin, Michael J. Black, and **Javier Romero**. SMPLpix: Neural avatars from 3D human models. In *Winter Conference on Applications of Computer Vision (WACV)*, pages 1810–1819, January 2021.

David Smith, Matthew Loper, Xiaochen Hu, Paris Mavroidis, and **Javier Romero**. Facsimile: Fast and accurate scans from an image in less than a second. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2019.

Gül Varol, **Javier Romero**, Xavier Martin, Naureen Mahmood, Michael J. Black, Ivan Laptev, and Cordelia Schmid. Learning from synthetic humans. In *Proceedings IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2017, Piscataway, NJ, USA, July 2017*. IEEE.

Javier Romero, Dimitrios Tzionas, and Michael J. Black. Embodied hands: Modeling and capturing hands and bodies together. *ACM Transactions on Graphics, (Proc. SIGGRAPH Asia)*, 36(6):245:1–245:17, November 2017.

Tianye Li, Timo Bolkart, Michael J. Black, Hao Li, and **Javier Romero**. Learning a model of facial shape and expression from 4D scans. *ACM Transactions on Graphics*, 36(6):194:1–194:17, November 2017. Two first authors contributed equally.

Thomas Feix, **Javier Romero**, Heinz-Bodo Schmiedmayer, Aaron M. Dollar, and Danica Kragic. The grasp taxonomy of human grasp types. *Human-Machine Systems, IEEE Transactions on*, 46(1):66–77, 2016.

Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, **Javier Romero**, and Michael J. Black. Keep it SMPL: Automatic estimation of 3D human pose and shape from a single image. In *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, pages 561–578. Springer International Publishing, October 2016.

Matthew Loper, Naureen Mahmood, **Javier Romero**, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM Trans. Graphics (Proc. SIGGRAPH Asia)*, 34(6):248:1–248:16, October 2015.

External Reviewer

Associate Editor in ICRA and IROS

Reviewer in journals CVIU, IJCV, SMC, PAMI, TRO, UAIS, JMIV, BB

Reviewer in conferences ICCV, CVPR, ECCV, NeurIPS, ICLR, ICRA, IROS, SIGGRAPH Asia, SIGGRAPH

Full list of publications (also available in [Google Scholar](#))

Journal Papers , ,

Chengan He, Junxuan Li, Tobias Kirschstein, Artem Sevastopolsky, Shunsuke Saito, Qingyang Tan, **Javier Romero**, Chen Cao, Holly E. Rushmeier, and Giljoo Nam. 3dgh: 3d head generation with composable hair and face. *ACM Trans. Graph.*, 44(4):59:1–59:12, 2025.

Feisal Rasras, Stanislav Pidhorskyi, Tomas Simon, Hallison Paz, He Wen, Jason M. Saragih, and **Javier Romero**. The lips, the teeth, the tip of the tongue: LTT tracking. In Takeo Igarashi, Ariel Shamir, and Hao (Richard) Zhang, editors, *SIGGRAPH Asia 2024 Conference Papers, SA 2024, Tokyo, Japan, December 3-6, 2024*, pages 115:1–115:11. ACM, 2024.

Donglai Xiang, Timur Bagautdinov, Tuur Stuyck, Fabian Prada, **Javier Romero**, Weipeng Xu, Shunsuke Saito, Jingfan Guo, Breannan Smith, Takaaki Shiratori, et al. Dressing avatars: Deep photorealistic appearance for physically simulated clothing. *ACM Transactions on Graphics (TOG)*, 41(6):1–15, 2022.

Anurag Ranjan, David T Hoffmann, Dimitrios Tzionas, Siyu Tang, **Javier Romero**, and Michael J Black. Learning multi-human optical flow. *International Journal of Computer Vision (IJCV)*, (128):873–890, April 2020.

Celia Foster, Mintao Zhao, **Javier Romero**, Michael J. Black, Betty J. Mohler, Andreas Bartels, and Isabelle Bühlhoff. Decoding subcategories of human bodies from both body- and face-responsive cortical regions. *NeuroImage*, 202(15):116085, November 2019.

Anne Thaler, Ivelina Piryanova, Jeanine K. Stefanucci, Sergi Pujades, Stephan de la Rosa, Stephan Streuber, **Javier Romero**, Michael J. Black, and Betty J. Mohler. Visual perception and evaluation of photo-realistic self-avatars from 3D body scans in males and females. *Frontiers in ICT*, 5:1–14, September 2018.

Mazen Al Borno, Ludovic Righetti, Michael J. Black, Scott L. Delp, Eugene Fiume, and **Javier Romero**. Robust physics-based motion retargeting with realistic body shapes. *Computer Graphics Forum*, 37:6:1–12, July 2018.

Simone Claire Mölbert, Anne Thaler, Betty J. Mohler, Stephan Streuber, **Javier Romero**, Michael J. Black, Stephan Zipfel, Hans-Otto Karnath, and Katrin Elisabeth Giel. Assessing body image in anorexia nervosa using biometric self-avatars in virtual reality: Attitudinal components rather than visual body size estimation are distorted. *Psychological Medicine*, 48(4):642–653, March 2018.

Anne Thaler, Michael N. Geuss, Simone C. Mölbert, Katrin E. Giel, Stephan Streuber, **Javier Romero**, Michael J. Black, and Betty J. Mohler. Body size estimation of self and others in females varying in BMI. *PLoS ONE*, 13(2), February 2018.

Tianye Li, Timo Bolkart, Michael J. Black, Hao Li, and **Javier Romero**. Learning a model of facial shape and expression from 4D scans. *ACM Transactions on Graphics*, 36(6):194:1–194:17, November 2017. Two first authors contributed equally.

Javier Romero, Dimitrios Tzionas, and Michael J. Black. Embodied hands: Modeling and capturing hands and bodies together. *ACM Transactions on Graphics, (Proc. SIGGRAPH Asia)*, 36(6):245:1–245:17, November 2017.

Si Y. Yeo, **Javier Romero**, Matthew Loper, Jürgen Machann, and Michael J. Black. Shape estimation of subcutaneous adipose tissue using an articulated statistical shape model. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging and Visualization*, 0(0):1–8, 2016.

Thomas Feix, **Javier Romero**, Heinz-Bodo Schmiedmayer, Aaron M. Dollar, and Danica Kragic. The grasp taxonomy of human grasp types. *Human-Machine Systems, IEEE Transactions on*, 46(1):66–77, 2016.

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Gerard Pons-Moll, **Javier Romero**, Naureen Mahmood, and Michael J. Black. Dyna: A model of dynamic human shape in motion. *ACM Transactions on Graphics, (Proc. SIGGRAPH)*, 34(4):120:1–120:14, August 2015.

Ivelina Piryankova, Jeanine Stefanucci, **Javier Romero**, Stephan de la Rosa, Michael J. Black, and Betty Mohler. Can i recognize my body’s weight? the influence of shape and texture on the perception of self. *ACM Transactions on Applied Perception for the Symposium on Applied Perception*, 11(3):13:1–13:18, September 2014.

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Thomas Feix, **Javier Romero**, Carl H. Ek, Heinz-Bodo Schmiedmayer, and Danica Kragic. A metric for comparing the anthropomorphic motion capability of artificial hands. *IEEE RAS Transactions on Robotics, TRO*, pages 974–980, 2012.

Xavier Gratal, **Javier Romero**, Jeannette Bohg, and Danica Kragic. Visual servoing on unknown objects. In *International Federation of Automatic Control, IFAC Mechatronics: The Science of Intelligent Machines*, pages 423–435, 2011.

Hedvig Kjellström, **Javier Romero**, and Danica Kragic. Visual object-action recognition: Inferring object affordances from human demonstration. *Computer Vision and Image Understanding*, pages 81–90, 2010.

Hedvig Kjellström, **Javier Romero**, Danica Martinez, and Danica Kragic. Simultaneous visual recognition of manipulation actions and manipulated objects. In *European Conference on Computer Vision, ECCV*, pages 336–349, 2008.

Conference Papers

Yuxuan Xue, Ruofan Liang, Egor Zakharov, Timur M. Bagautdinov, Chen Cao, Giljoo Nam, Shunsuke Saito, Gerard Pons-Moll, and **Javier Romero**. Georelight: Learning joint geometrical relighting and reconstruction with flexible multi-modal diffusion transformers. In *IEEE/CVF Conference on Computer*

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Jinhyung Park, **Javier Romero**, Shunsuke Saito, Fabian Prada, Takaaki Shiratori, Yichen Xu, Federica Bogó, Shouo-I Yu, Kris Kitani, and Rawal Khirodkar. ATLAS: decoupling skeletal and shape parameters for expressive parametric human modeling. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 6508–6518. IEEE, 2025.

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Shaofei Wang, Tomas Simon, Igor Santesteban, Timur M. Bagautdinov, Junxuan Li, Vasu Agrawal, Fabian Prada, Shouo-I Yu, Pace Nalbene, Matt Gramlich, Roman Lubachersky, Chenglei Wu, **Javier Romero**, Jason M. Saragih, Michael Zollhöfer, Andreas Geiger, Siyu Tang, and Shunsuke Saito. Relightable full-body gaussian codec avatars. In Ginger Alford, Hao (Richard) Zhang, and Adriana Schulz, editors, *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference, SIGGRAPH Conference Papers 2025, Vancouver, BC, Canada, August 10-14, 2025*, pages 146:1–146:12. ACM, 2025.

Rong Wang, Fabian Prada, Ziyang Wang, Zhongshi Jiang, Chengxiang Yin, Junxuan Li, Shunsuke Saito, Igor Santesteban, **Javier Romero**, Rohan Joshi, Hongdong Li, Jason M. Saragih, and Yaser Sheikh. FRESA: feedforward reconstruction of personalized skinned avatars from few images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 281–291. Computer Vision Foundation / IEEE, 2025.

Julieta Martinez, Emily Kim, **Javier Romero**, Timur M. Bagautdinov, Shunsuke Saito, Shouo-I Yu, Stuart Anderson, Michael Zollhöfer, Te-Li Wang, Shaojie Bai, Chenghui Li, Shih-En Wei, Rohan Joshi, Wyatt Borsos, Tomas Simon, Jason M. Saragih, Paul Theodosis, Alexander Greene, Anjani Josyula, Silvio Maeta, Andrew Jewett, Simion Venshtain, Christopher Heilman, Yueh-Tung Chen, Sidi Fu, Mohamed Elshaer, Tingfang Du, Longhua Wu, Shen-Chi Chen, Kai Kang, Michael Wu, Youssef Emad, Steven Longay, Ashley Brewer, Hitesh Shah, James Booth, Taylor Koska, Kayla Haidle, Matthew Andromalos, Joanna Hsu, Thomas Dauer, Peter Selednik, Timothy Godisart, Scott Ardisson, Matthew Cipperly, Ben Humberston, Lon Farr, Bob Hansen, Peihong Guo, Dave Braun, Steven Krenn, He Wen, Lucas Evans, Natalia Fadeeva, Matthew Stewart, Gabriel Schwartz, Divam Gupta, Gyeongsik Moon, Kaiwen Guo, Yuan Dong, Yichen Xu, Takaaki Shiratori, Fabian Prada, Bernardo Pires, Bo Peng, Julia Buffalini, Autumn Trimble, Kevyn McPhail, Melissa Schoeller, and Yaser Sheikh. Codec avatar studio: Paired human captures for complete, driveable, and generalizable avatars. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang, editors, *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024.

Jingfan Guo, Fabian Prada, Donglai Xiang, **Javier Romero**, Chenglei Wu, Hyun Soo Park, Takaaki Shiratori, and Shunsuke Saito. Diffusion shape prior for wrinkle-accurate cloth registration. In *2024 International Conference on 3D Vision (3DV)*, pages 790–799. IEEE, 2024.

Zhaoxi Chen, Gyeongsik Moon, Kaiwen Guo, Chen Cao, Stanislav Pidhorskyi, Tomas Simon, Rohan Joshi, Yuan Dong, Yichen Xu, Bernardo Pires, He Wen, Lucas Evans, Bo Peng, Julia Buffalini, Autumn Trimble, Kevyn McPhail, Melissa Schoeller, Shou-I Yu, **Javier Romero**, Michael Zollhoefer, Yaser Sheikh, Ziwei Lie, and Shunsuke Saito. Urhand: Universal relightable hands. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 119–129, 2024.

Evonne Ng, **Javier Romero**, Timur Bagautdinov, Shaojie Bai, Trevor Darrell, Angjoo Kanazawa, and Alexander Richard. From audio to photoreal embodiment: Synthesizing humans in conversations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1001–1010, 2024.

Wojciech Zielonka, Timur M. Bagautdinov, Shunsuke Saito, Michael Zollhöfer, Justus Thies, and **Javier Romero**. Drivable 3d gaussian avatars. In *International Conference on 3D Vision, 3DV 2025, Singapore, March 25-28, 2025*, pages 979–990. IEEE, 2025.

Ziqian Bai, Timur Bagautdinov, **Javier Romero**, Michael Zollhöfer, Ping Tan, and Shunsuke Saito. Autoavatar: Autoregressive neural fields for dynamic avatar modeling. In *European Conference on Computer Vision*, pages 222–239. Springer, 2022.

Nataniel Ruiz, Miriam Bellver, Timo Bolkart, Ambuj Arora, Ming C Lin, **Javier Romero**, and Raja Bala. Human body measurement estimation with adversarial augmentation. In *2022 International Conference on 3D Vision (3DV)*, pages 219–230. IEEE, 2022.

Soubhik Sanyal, Alex Vorobiov, Timo Bolkart, Matthew Loper, Betty Mohler, Larry S Davis, **Javier Romero**, and Michael J Black. Learning realistic human posing using cyclic self-supervision with 3d shape, pose, and appearance consistency. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11138–11147, 2021.

Sergey Prokudin, Michael J. Black, and **Javier Romero**. SMPLpix: Neural avatars from 3D human models. In *Winter Conference on Applications of Computer Vision (WACV)*, pages 1810–1819, January 2021.

David Smith, Matthew Loper, Xiaochen Hu, Paris Mavroidis, and **Javier Romero**. Facsimile: Fast and accurate scans from an image in less than a second. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2019.

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Anurag Ranjan, **Javier Romero**, and Michael J. Black. Learning human optical flow. In *29th British Machine Vision Conference*, September 2018.

Nikolas Hesse, Sergi Pujades, **Javier Romero**, Michael J. Black, Christoph Bodensteiner, Michael Arens, Ulrich G. Hofmann, Uta Tacke, Mijna Hadders-Algra, Raphael Weinberger, Wolfgang Muller-Felber, and A. Sebastian Schroeder. Learning an infant body model from RGB-D data for accurate full body motion analysis. In *Int. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, September 2018.

Julietta Martinez, Rayat Hossain, **Javier Romero**, and James J. Little. A simple yet effective baseline for 3d human pose estimation. In *Proceedings IEEE International Conference on Computer Vision (ICCV)*, Piscataway, NJ, USA, October 2017. IEEE.

Lukas Balles, **Javier Romero**, and Philipp Hennig. Coupling adaptive batch sizes with learning rates. In *Proceedings Conference on Uncertainty in Artificial Intelligence (UAI) 2017*, pages 410–419. Association for Uncertainty in Artificial Intelligence (AUAI), August 2017.

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Gül Varol, **Javier Romero**, Xavier Martin, Naureen Mahmood, Michael J. Black, Ivan Laptev, and Cordelia Schmid. Learning from synthetic humans. In *Proceedings IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2017*, Piscataway, NJ, USA, July 2017. IEEE.

Julietta Martinez, Michael J. Black, and **Javier Romero**. On human motion prediction using recurrent neural networks. In *Proceedings IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2017*, Piscataway, NJ, USA, July 2017. IEEE.

Christoph Lassner, **Javier Romero**, Martin Kiefel, Federica Bogo, Michael J. Black, and Peter V. Gehler. Unite the people: Closing the loop between 3d and 2d human representations. In *Proceedings IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2017*, Piscataway, NJ, USA, July 2017. IEEE.

Reuben Fleming, Betty J. Mohler, **Javier Romero**, Michael J. Black, and Martin Breidt. *Appealing Avatars from 3D Body Scans: Perceptual Effects of Stylization*, pages 175–196. Springer International Publishing, 2017.

Yinghao Huang, Federica Bogo, Christoph Lassner, Angjoo Kanazawa, Peter V. Gehler, **Javier Romero**, Ijaz Akhter, and Michael J. Black. Towards accurate marker-less human shape and pose estimation over time. In *International Conference on 3D Vision (3DV)*, pages 421–430, 2017.

Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, **Javier Romero**, and Michael J. Black. Keep it SMPL: Automatic estimation of 3D human pose and shape from a single image. In *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, pages 561–578. Springer International Publishing, October 2016.

Reuben Fleming, Betty Mohler, **Javier Romero**, Michael J. Black, and Martin Breidt. Appealing female avatars from 3D body scans: Perceptual effects of stylization. In *11th Int. Conf. on Computer Graphics Theory and Applications (GRAPP)*, February 2016.

Federica Bogo, Michael J. Black, Matthew Loper, and **Javier Romero**. Detailed full-body reconstructions of moving people from monocular RGB-D sequences. In *International Conference on Computer Vision (ICCV)*, pages 2300–2308, December 2015.

Javier Romero, Matthew Loper, and Michael J. Black. Flowcap: 2d human pose from optical flow. In *German Conference on Pattern Recognition (GCPR)*, 2015.

Federica Bogo, **Javier Romero**, Matthew Loper, and Michael J. Black. FAUST: Dataset and evaluation for 3D mesh registration. In *Proceedings IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, pages 3794–3801, Columbus, Ohio, USA, June 2014.

Federica Bogo, **Javier Romero**, Enoch Peserico, and Michael J. Black. Automated detection of new or evolving melanocytic lesions using a 3D body model. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, volume 8673 of *Lecture Notes in Computer Science*, pages 593–600. Springer International Publishing, September 2014.

Jeannette Bohg, **Javier Romero**, Alexander Herzog, and Stefan Schaal. Robot arm pose estimation through pixel-wise part classification. In *IEEE International Conference on Robotics and Automation (ICRA) 2014*, pages 3143–3150, June 2014.

Silvia Zuffi, **Javier Romero**, Cordelia Schmid, and Michael J. Black. Estimating human pose with flowing puppets. In *IEEE International Conference on Computer Vision (ICCV)*, pages 3312–3319, 2013.

Xavier Gratal, **Javier Romero**, and Danica Kragic. Virtual visual servoing for real-time robot pose estimation. In *International Federation of Automatic Control World Congress, IFAC*, 2011.

Javier Romero, Thomas Feix, Heinz-Bodo Kjellström, and Danica Kragic. Spatio-temporal modeling of grasping actions. In *IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS*, pages 2103–2108, 2010.

Javier Romero, Hedvig Kjellström, and Danica Kragic. Hands in action: real-time 3D reconstruction of hands in interaction with objects. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 458–463, 2010.

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Martin Do, **Javier Romero**, Hedvig Kjellström, Pedram Azad, Tamim Asfour, Danica Kragic, and Rudiger Dillmann. Grasp recognition and mapping on humanoid robots. In *IEEE-RAS International Conference on Humanoid Robots*, pages 465–471, 2009.

Javier Romero, Hedvig Kjellström, and Danica Kragic. Modeling and evaluation of human-to-robot mapping of grasps. In *International Conference on Advanced Robotics (ICAR)*, pages 1–6, 2009.

Javier Romero, Hedvig Kjellström, and Danica Kragic. Monocular real-time 3d articulated hand pose estimation. In *IEEE-RAS International Conference on Humanoid Robots*, pages 87–92, 2009.

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